

MRMhub Data Processing Workflow for Dataset 4

Supplementary Code 3

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1 Overview

Dataset 4 is a **fully quantitative steroid panel** used to validate MRMhub against a vendor pipeline (Agilent MassHunter). Fifteen steroids were measured in human serum with the IBL International *Steroid Panel LC-MS* kit (cat. 30191875) according to the instructions for use, with modifications. The kit acquires its analytes over two chromatographic panels; **only Panel 1 (15 analytes) is used here**, which is why every sample identifier carries the `_P1` suffix (e.g. `Cal_A_P1`, `QC_High1_P1`).

Unlike Datasets 1 and 3 (relative lipidomics), this is an externally calibrated absolute assay. The analytical design comprises an **external calibration curve** (`Cal_0–Cal_F`), **low and high quality-control samples** (`QC_Low1`, `QC_High1`), and **external quality-assessment (EQA) samples** — SKML ring-trial materials with reference-method-assigned target concentrations (Jansen et al. 2014).

The workflow compares MRMhub and MassHunter at both pipeline stages:

- **INTEGRATOR** — peak areas from MRMhub-INTEGRATOR vs a MassHunter re-integration of the same raw files (Figure 7).
- **QUANT** — calibration fits and concentrations, ending in accuracy against the SKML EQA reference values (Figure 12).

2 Raw Data Processing: Peak Picking and Integration

Raw `.d` files were converted to `mzML` and integrated with MRMhub-INTEGRATOR, as described for Datasets 1 and 3. The batch was integrated in two subsets — the **system-suitability** injections (`SST`, ten replicate injections) and the **assay** samples (`ASSAY`: blanks, calibrators, QCs, EQA samples) — exported as separate long-format tables and combined downstream. INTEGRATOR reports peak areas, retention times and peak metadata in `long.csv`, and writes per-transition integration plots (the shaded, integrated region) to the `by_transition` folder.

The same raw files were independently re-integrated in Agilent MassHunter Quantitative Analysis, providing the reference peak areas and concentrations used throughout this document.

Note

The example chromatograms are cropped from the INTEGRATOR `by_transition` PDFs by `figures/dataset4/dataset4-chromatograms.R` (run once; requires the `magick` R package and `ghostscript`) and embedded here as static images from `images/dataset4/`. The notebook render itself needs neither.

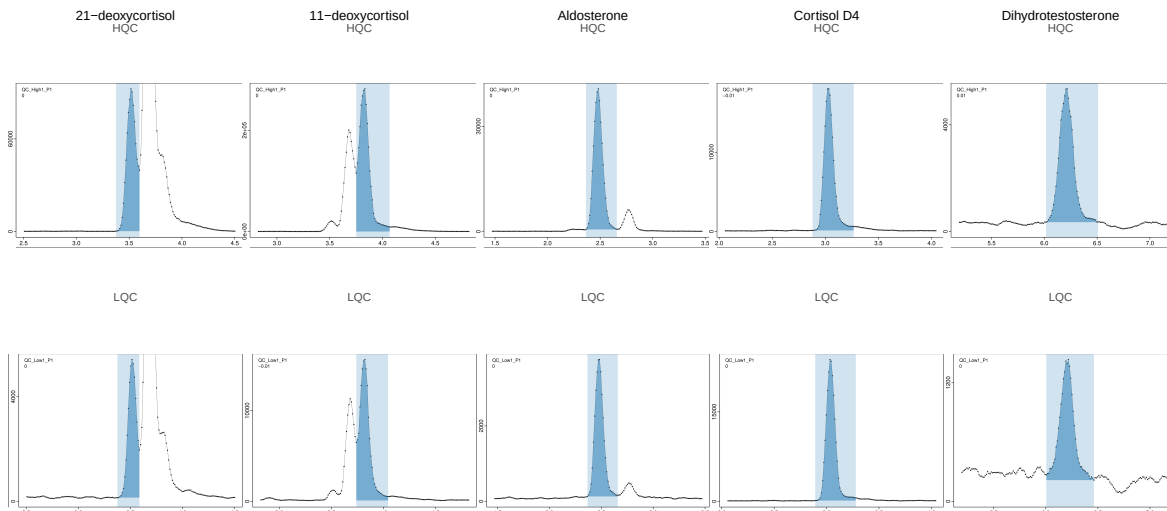


Figure 1: **Example INTEGRATOR chromatograms.** MRMhub-INTEGRATOR chromatograms for 21-deoxycortisol, 11-deoxycortisol, Aldosterone, Cortisol D4 (internal standard) and Dihydrotestosterone at high (HQC) and low (LQC) QC levels; the shaded region is the integrated peak.

3 Data Postprocessing and QC

The postprocessing pipeline is the QUANT module (`library(mrmhub)`): import the INTEGRATOR results and metadata, normalize to internal standards, fit external calibration curves, and quantify. This is a shorter pipeline than Datasets 1/3 — no drift/batch correction or feature filtering, which do not apply to a 15-analyte, single-batch panel.

i mrmhub version

This notebook was rendered with **mrmhub 0.9.3** (QUANT module of **MRMhub**). This API is currently on the `development` branch; install with `remotes::install_github("SLINGhub/MRMhub", ref = "development")`.

```
library(mrmhub)
library(mirai) # parallel processing (picked up automatically by mrmhub)
#>
#> Attaching package: 'mirai'
#> The following object is masked from 'package:mrmhub':
#>
#>     status

# Set number of cores for parallel processing (all-but-one, so the system stays responsive)
```

```

n_cores <- {
  n <- parallel::detectCores()
  if (is.na(n)) n <- 4L
  max(1L, n - 1L)
}
if (mirai::status()$daemons == 0) mirai::daemons(n_cores)

```

3.1 Import INTEGRATOR results and metadata

The two INTEGRATOR subsets share columns and hold disjoint analyses, so they are combined by row-binding. The combined table is written to `output/` because it is re-imported later (Figure 12) with MassHunter areas substituted in.

```

sst_long <- read_csv(
  file.path(d4, "Dataset4_MRMhub-INTEGRATOR_SST.csv"),
  show_col_types = FALSE)
assay_long <- read_csv(
  file.path(d4, "Dataset4_MRMhub-INTEGRATOR_ASSAY.csv"),
  show_col_types = FALSE)
long_combined <- bind_rows(sst_long, assay_long)

long_combined_path <- file.path("output", "Dataset4_long_combined.csv")
write_csv(long_combined, long_combined_path)

# Analysis ids per subset (import strips the ".mzML" extension).
sst_analysis_ids <- unique(str_remove(
  sst_long$raw_data_filename,
  "[.]mzML$"))
assay_analysis_ids <- unique(str_remove(
  assay_long$raw_data_filename,
  "[.]mzML$"))

meta_path <- file.path(d4, "Dataset4_Metadata.xlsx")

mexp <- MRHubExperiment(title = "Dataset 4 - Steroid Assay")
mexp <- import_data_mrmhub(
  mexp,
  path = long_combined_path,
  import_metadata = TRUE)
#> v Imported 30 analyses with 64 features.
#> i feature_area selected as default feature intensity. Modify with `set_intensity_var()`.

```

```

#> Warning: ! Unrecognized qc_type value(s): "BLK".
#> i These are not standard QC types and may be dropped from QC metrics and plots
#> that expect them.
#> i Standard values: "SBLK", "TBLK", "UBLK", "HQC", "MQC", "LQC", "QC", "PBLK",
#> "CAL", "EQA", "PQC", "TQC", "BQC", "RQC", "EQC", "NIST", "LTR", "SPL", "SST",
#> and "MBLK" ("Sample" is an alias for "SPL").
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 64 features.
mexp <- import_metadata_msorganiser(
  mexp,
  path = meta_path,
  excl_unmatched_analyses = TRUE, ignore_warnings = TRUE)
#> Found no errors, 2 warnings, and 2 notes in the metadata.
#> -----
#>   Type   Table      Column      Issue                                     Count
#> 1 W*     Analyses analysis_id Analyses not in analysis data             2
#> 2 W*     Features feature_id  Feature(s) without metadata        34
#> 3 N       Analyses sample_id   Not defined for all analyses      16
#> 4 N       Features analyte_id  Not defined for all features       15
#>
#> -----
#> E = Error, W = Warning, W* = Suppressed Warning, N = Note
#> -----
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
#> v Internal Standard metadata associated with 15 ISTDs.
#> v QC concentration metadata associated with 13 annotated samples and 15 annotated analytes.

```

3.2 Normalization and calibration

Each feature is normalized to its stable-isotope-labelled internal standard, then a quadratic, 1/x-weighted external calibration curve is fitted per analyte from the Cal₀–Cal_F levels.

```

mexp <- normalize_by_istd(mexp)
#> v 15 features normalized with 15 ISTDs in 30 analyses.
mexp <- calc_calibration_results(
  mexp,
  fit_overwrite = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x")
#> v Calibration curve fits calculated for all 15 quantifier features. Average r-squared: 0.9

```

The fitted curves and their metrics are inspected before quantification.

```
plot_calibrationcurves(
  mexp,
  fit_overwrite = FALSE,
  fit_model = "quadratic",
  fit_weighting = "1/x",
  include_istd = FALSE,
  include_qualifier = FALSE,
  show_progress = FALSE)
#> v Calibration curve fits calculated for all 15 quantifier features. Average r-squared: 0.9992.
#> i Generating plots (1 page)...
#> v Done
```

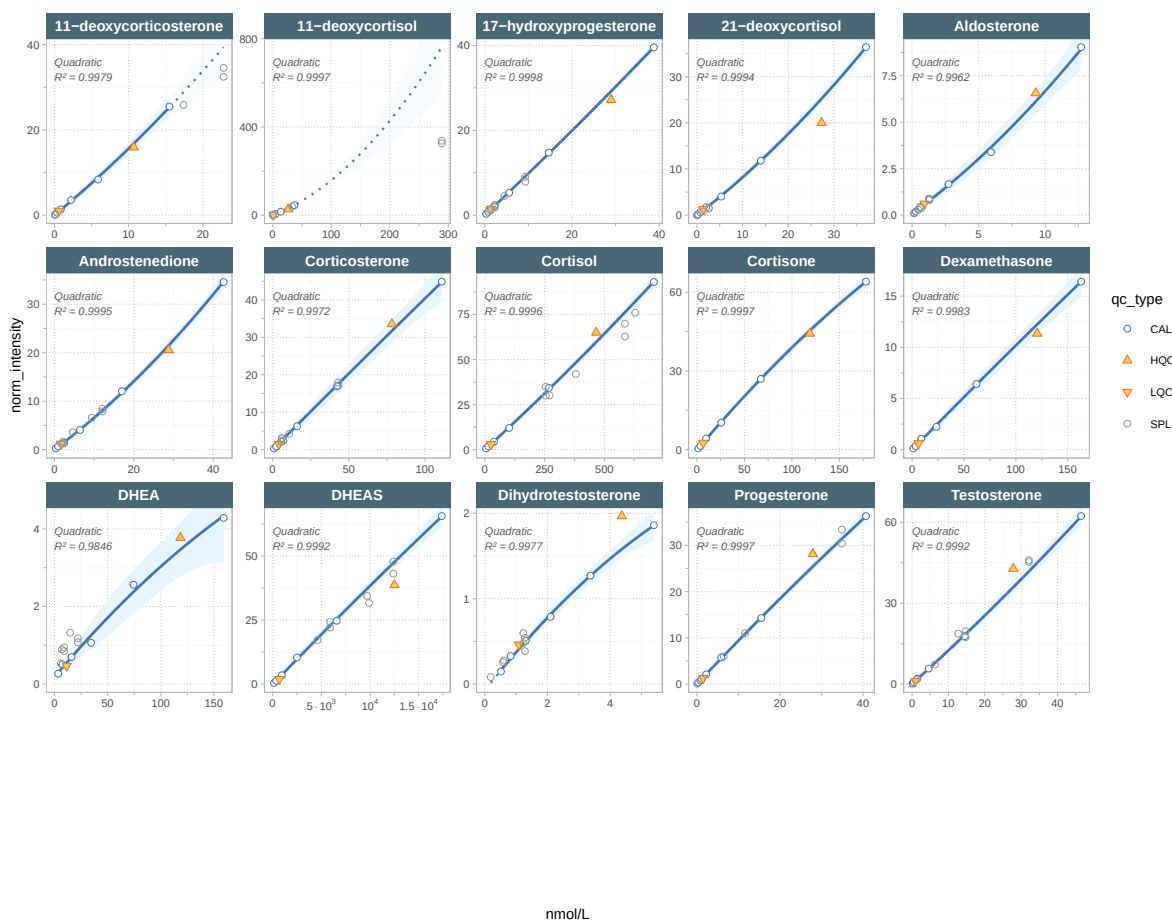


Figure 2: **External calibration curves.** Quadratic, 1/x-weighted curves for the 15 steroid analytes (quantifier transitions).

Table 1: **Calibration metrics per analyte.** Fit model, weighting, R^2 , LOD and LOQ for each quantifier.

```
get_calibration_metrics(mexp) |>
  filter(is_quantifier) |>
  transmute(
    analyte = feature_id,
    fit_model, fit_weighting, r2 = round(r2, 5), lod = signif(lod, 3),
    loq = signif(loq, 3))
#> # A tibble: 15 x 6
#>   analyte          fit_model fit_weighting    r2    lod    loq
#>   <chr>          <chr>      <chr>      <dbl> <dbl> <dbl>
#> 1 11-deoxycorticosterone quadratic 1/x      0.998 0.457 1.38
#> 2 11-deoxycortisol      quadratic 1/x      1.000 0.279 0.844
#> 3 17-hydroxyprogesterone quadratic 1/x      1.000 0.197 0.596
#> 4 21-deoxycortisol      quadratic 1/x      0.999 0.427 1.29
#> 5 Aldosterone          quadratic 1/x      0.996 0.639 1.94
#> 6 Androstenedione       quadratic 1/x      0.999 0.445 1.35
#> 7 Corticosterone        quadratic 1/x      0.997 1.25  3.79
#> 8 Cortisol              quadratic 1/x      1.000 1.22  3.71
#> 9 Cortisone             quadratic 1/x      1.000 0.463 1.4
#> 10 DHEA                 quadratic 1/x      0.985 3.03  9.18
#> 11 DHEAS                quadratic 1/x      0.999 8.29 25.1
#> 12 Dexamethasone        quadratic 1/x      0.998 1.19  3.6
#> 13 Dihydrotestosterone  quadratic 1/x      0.998 0.178 0.54
#> 14 Progesterone         quadratic 1/x      1.000 0.251 0.76
#> 15 Testosterone         quadratic 1/x      0.999 0.458 1.39
```


3.3 Quantification

Concentrations are computed from the normalized intensities via the calibration curves. Qualifier transitions are excluded; analytes whose curve failed to fit are skipped rather than aborting the run.

```
mexp <- quantify_by_calibration(
  mexp,
  fit_overwrite = FALSE,
  include_qualifier = FALSE,
  ignore_failed_calibration = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x")
#> v Calibration curve fits calculated for all 15 quantifier features. Average r-squared: 0.9
#> i 77 concentration values fall outside the calibrated range (retained, flagged in feature)
#> v Concentrations calculated for 15 features in 30 analyses.
#> v Concentrations are given in nmol/L.

# Export + reload the concentration table (wide: analysis_id, qc_type, analytes).
conc_path <- file.path("output", "Dataset4_quant_results.csv")
save_dataset_csv(
  mexp,
  path = conc_path,
  variable = "conc",
  add_qctype = TRUE)
#> v Concentration values for 30 analyses and 15 features have been exported to 'output/Dataset4_quant_results.csv'
d_conc_raw <- read_csv(conc_path, show_col_types = FALSE)
```

3.4 MassHunter re-integration

The MassHunter Quantitative Analysis export is imported as a second experiment and reshaped to one tidy table — the single MassHunter source for all comparisons below. The trailing (ISTD) suffix on labelled internal standards is stripped so feature identifiers match MRMhub.

```
mh_reint <- import_data_masshunter(
  MRMhubExperiment(title = "MassHunter re-integration"),
  path = file.path(
    d4,
    "Dataset4_IntegrationResults_MassHunter_IsoSP-22_V3.csv"),
  import_metadata = TRUE, silent = TRUE)
```

```
#> i feature_area selected as default feature intensity. Modify with `set_intensity_var()`.
#> Warning: ! Unrecognized qc_type value(s): "ResponseCheck", "DoubleBlank", "Blank",
#>   "MatrixBlank", and "Cal".
#> i These are not standard QC types and may be dropped from QC metrics and plots
#>   that expect them.
#> i Standard values: "SBLK", "TBLK", "UBLK", "HQC", "MQC", "LQC", "QC", "PBLK",
#>   "CAL", "EQA", "PQC", "TQC", "BQC", "RQC", "EQC", "NIST", "LTR", "SPL", "SST",
#>   and "MBLK" ("Sample" is an alias for "SPL").
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
mh_data <- get_analyticaldata(mh_reint, annotated = FALSE) |>
  transmute(
    analysis_id,
    feature_id = str_remove(feature_id, "[ ]*\\(ISTD\\)$"),
    rt_mh = feature_rt, area_mh = feature_area,
    conc_mh = feature_conc_final, mi_mh = feature_manual_integration)
```

3.5 Below-Cal-A exclusion for study samples

Study samples (SPL = the SKML EQA samples) whose **quantifier peak area falls below the corresponding Cal A area** are below the lowest non-zero calibrator and treated as not reliably quantifiable. The rule is applied per analyte and **independently per method** — MRMhub SPL area vs MRMhub Cal A area, MassHunter SPL area vs MassHunter Cal A area — yielding the flags `excl_mrm` / `excl_mh` that downstream comparisons apply to their SPL rows. Internal standards are never filtered.

```
CAL_A_ID <- "Cal A_P1"

# MRMhub quantifier areas (quantifiers only, excluding the ISTDs).
mrm_q_area <- get_analyticaldata(mexp, annotated = TRUE) |>
  filter(is_quantifier, !is_istd) |>
  transmute(
    analysis_id,
    analyte_id = feature_id, qc_type, area_mrm = feature_area)

cala_mrm <- mrm_q_area |>
  filter(analysis_id == CAL_A_ID) |>
  select(analyte_id, cala_area_mrm = area_mrm)
cala_mh <- mh_data |>
  filter(analysis_id == CAL_A_ID) |>
  transmute(analyte_id = feature_id, cala_area_mh = area_mh)
```

Table 2: **Low/high QC concentrations.** Quantified concentrations of the low and high QC samples, one row per analyte.

```
d_conc_raw |>
  filter(qc_type == "QC") |>
  pivot_longer(
    !c(analysis_id, qc_type),
    names_to = "analyte", values_to = "conc") |>
  mutate(
    qc = if_else(str_detect(analysis_id, "High"), "QC_High", "QC_Low")) |>
  select(analyte, qc, conc) |>
  pivot_wider(names_from = qc, values_from = conc) |>
  mutate(across(where(is.numeric), ~ signif(.x, 3)))
#> # A tibble: 0 x 1
#> # i 1 variable: analyte <chr>
```

```
# One row per SPL (analysis, analyte); flag below-Cal-A per method (NA-safe).
spl_area_excl <- mrm_q_area |>
  filter(qc_type == "SPL") |>
  left_join(
    select(mh_data, analysis_id, analyte_id = feature_id, area_mh),
    by = c("analysis_id", "analyte_id")) |>
  left_join(cala_mrm, by = "analyte_id") |>
  left_join(cala_mh, by = "analyte_id") |>
  mutate(
    excl_mrm = !is.na(area_mrm) &
      !is.na(cala_area_mrm) &
      area_mrm < cala_area_mrm, excl_mh = !is.na(area_mh) &
      !is.na(cala_area_mh) &
      area_mh < cala_area_mh) |>
  select(
    analysis_id,
    analyte_id, area_mrm, area_mh, cala_area_mrm, cala_area_mh, excl_mrm,
    excl_mh)
```

3.6 Low / high QC

Quantified concentrations of the low and high QC samples, one row per analyte.

4 MRMhub vs MassHunter — Peak Integration

Figure 7 compares MRMhub-INTEGRATOR peak areas with the MassHunter re-integration of the same raw files. MRMhub quantifier and ISTD peaks are joined to MassHunter on `analysis_id + feature_id`, dropping near-noise peaks (area < 100), empty CAL rows, and below-Cal-A SPL points.

```
# Manuscript house theme + palettes (shared by both figures).
theme_set(
  theme_bw(base_size = 8, base_family = "sans") +
  theme(
    panel.grid.minor = element_blank(),
    legend.position = "bottom",
    plot.title = element_text(size = 8, face = "plain"),
    legend.key.size = unit(3.2, "mm"), legend.margin = margin(1, 1, 1, 1))

col_method <- c(MRMhub = "#2C7FB8", MassHunter = "#D95F0E")
smp_cols <- c(
  CAL = "#0072B2",
  HQC = "#D55E00", LQC = "#E69F00", SPL = "grey30", SST = "#009E73",
  SBLK = "#CC79A7", PBLK = "#999999")

# Save a composite as vector PDF + 300-dpi PNG at a given size (mm) for the manuscript.
save_fig <- function(plot, stem, height, width = 180) {
  ggsave(
    file.path("output", paste0(stem, ".pdf")),
    plot, device = cairo_pdf, width = width, height = height, units = "mm")
  ggsave(
    file.path("output", paste0(stem, ".png")),
    plot, device = ragg::agg_png, width = width, height = height,
    units = "mm", dpi = 300)
  invisible(plot)
}
```

```
mrm_peaks_q <- get_analyticaldata(mexp, annotated = TRUE) |>
  filter(is_quantifier | is_istd) |>
  transmute(
    analysis_id,
    feature_id, qc_type, sample_id,
    feature_type = factor(
      if_else(is_istd, "ISTD", "Quantifier"),
      levels = c("Quantifier", "ISTD")),
```

```

    rt_mrm = feature_rt, area_mrm = feature_area)

d_peakcmp_reint <- mrm_peaks_q |>
  inner_join(mh_data, by = c("analysis_id", "feature_id")) |>
  filter(
    !is.na(rt_mrm),
    !is.na(rt_mh), !is.na(area_mrm), !is.na(area_mh)) |>
  filter(area_mrm >= 100, area_mh >= 100) |>
  filter(!(qc_type == "CAL" & is.na(sample_id) & feature_type != "ISTD")) |>
  left_join(
    select(spl_area_excl, analysis_id, analyte_id, excl_mrm, excl_mh),
    by = c("analysis_id", "feature_id" = "analyte_id")) |>
  filter(
    !(qc_type == "SPL" &
      (coalesce(excl_mrm, FALSE) | coalesce(excl_mh, FALSE)))) |>
  select(-excl_mrm, -excl_mh) |>
  mutate(
    area_diff_pct = (area_mh - area_mrm) / area_mrm * 100,
    rt_diff = rt_mh - rt_mrm,
    subset = case_when(
      analysis_id %in% sst_analysis_ids ~ "SST",
      analysis_id %in% assay_analysis_ids ~ "ASSAY", TRUE ~ NA_character_))

```

4.1 Peak-area agreement

Log-log peak areas against the 1:1 line (Pearson r on log10 values). CAL/QC/SST samples span four orders of magnitude; agreement is essentially exact.

```

smp_lv <- c("CAL", "HQC", "LQC", "SST")
smp_cols4 <- c(
  CAL = "#0072B2",
  HQC = "#D55E00", LQC = "#E69F00", SST = "#009E73")
ft_shapes <- c(Quantifier = 16, ISTD = 17)
scale_smp <- scale_colour_manual(
  values = smp_cols4,
  limits = smp_lv, drop = FALSE)
scale_ft <- scale_shape_manual(values = ft_shapes)

d_cmp <- d_peakcmp_reint |>
  filter(qc_type %in% smp_lv) |>
  mutate(qc_type = factor(qc_type, levels = smp_lv))

```

```

r_log <- cor(log10(d_cmp$area_mrm), log10(d_cmp$area_mh))
ax_lims <- range(c(d_cmp$area_mrm, d_cmp$area_mh))
p_corr <- d_cmp |>
  ggplot(aes(area_mrm, area_mh)) +
  geom_abline(
    slope = 1,
    intercept = 0, linetype = "dashed", colour = "grey40") +
  geom_point(
    aes(colour = qc_type, shape = feature_type),
    size = 1.1, alpha = 0.7) +
  # Ring the peaks MassHunter integrated manually (MI flag), keeping the
  # sample-type colour / feature shape encoding underneath.
  geom_point(
    data = ~ filter(.x, mi_mh),
    aes(alpha = "Manual integration (MassHunter)"), shape = 1,
    colour = "grey15", size = 2.3, stroke = 0.5) +
  annotate(
    "text",
    x = ax_lims[1], y = ax_lims[2], hjust = 0, vjust = 1, size = 2.4,
    label = sprintf("Pearson r = %.4f", r_log)) +
  scale_x_log10(limits = ax_lims) +
  scale_y_log10(limits = ax_lims) +
  scale_smp +
  scale_ft +
  scale_alpha_manual(
    name = NULL,
    values = c("Manual integration (MassHunter)" = 1)) +
  guides(
    colour = guide_legend(order = 1, nrow = 2),
    shape = guide_legend(order = 2, nrow = 2),
    alpha = guide_legend(
      order = 3,
      override.aes = list(
        shape = 1,
        colour = "grey15", size = 2.3, stroke = 0.5))) +
  labs(
    x = "MRMhub peak area",
    y = "MassHunter peak area", colour = "Sample type", shape = "Feature",
    title = "Peak-area agreement")
p_corr

```

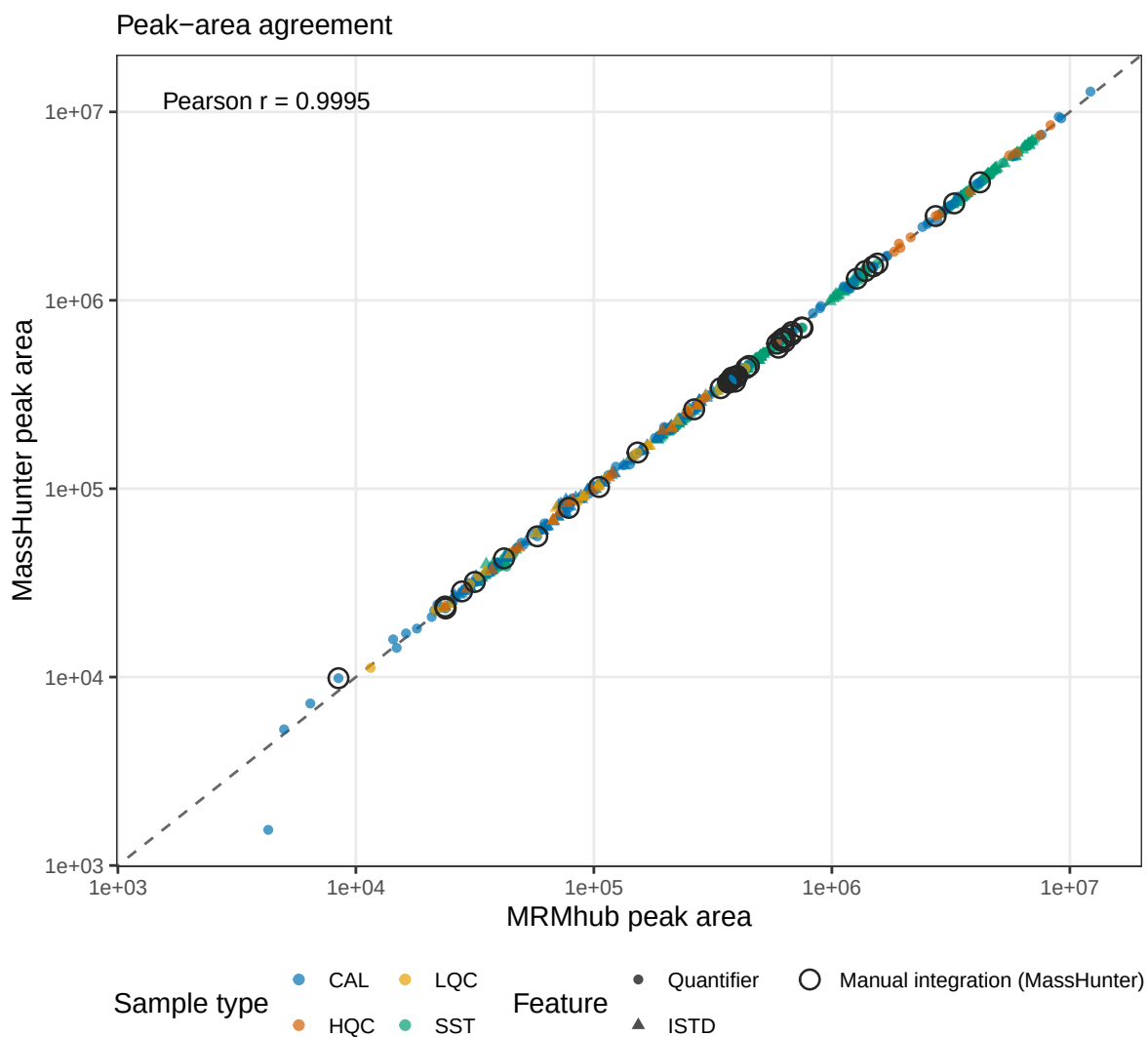


Figure 3: **Peak-area agreement.** MRMhub vs MassHunter peak areas (log-log) against the 1:1 line; Pearson r on log10 values across CAL/QC/SST samples. Open grey rings mark peaks MassHunter integrated manually. Panel A of Figure 7.

4.2 Precision (%CV) equivalence

Per-feature %CV over the ten replicate SST injections, MRMhub vs MassHunter. The two methods deliver equivalent precision (points on the 1:1 line).

```
sst_cv <- d_peakcmp_reint |>
  filter(subset == "SST") |>
  group_by(feature_type, feature_id) |>
```

```

summarise(
  cv_mrmhub = sd(area_mrm) / mean(area_mrm) * 100,
  cv_masshunter = sd(area_mh) / mean(area_mh) * 100, .groups = "drop")
cv_lims <- range(c(sst_cv$cv_mrmhub, sst_cv$cv_masshunter)) + c(-0.3, 0.3)
p_cv <- sst_cv |>
ggplot(aes(cv_mrmhub, cv_masshunter, shape = feature_type)) +
  geom_abline(
    slope = 1,
    intercept = 0, linetype = "dashed", colour = "grey60") +
  geom_point(size = 1.6, alpha = 0.85, colour = "#009E73") +
  coord_cartesian(xlim = cv_lims, ylim = cv_lims) +
  scale_ft +
  guides(shape = "none") +
  labs(
    x = "MRMhub %CV (10 SST reps)",
    y = "MassHunter %CV", title = "Precision (%CV) equivalence")
p_cv

```

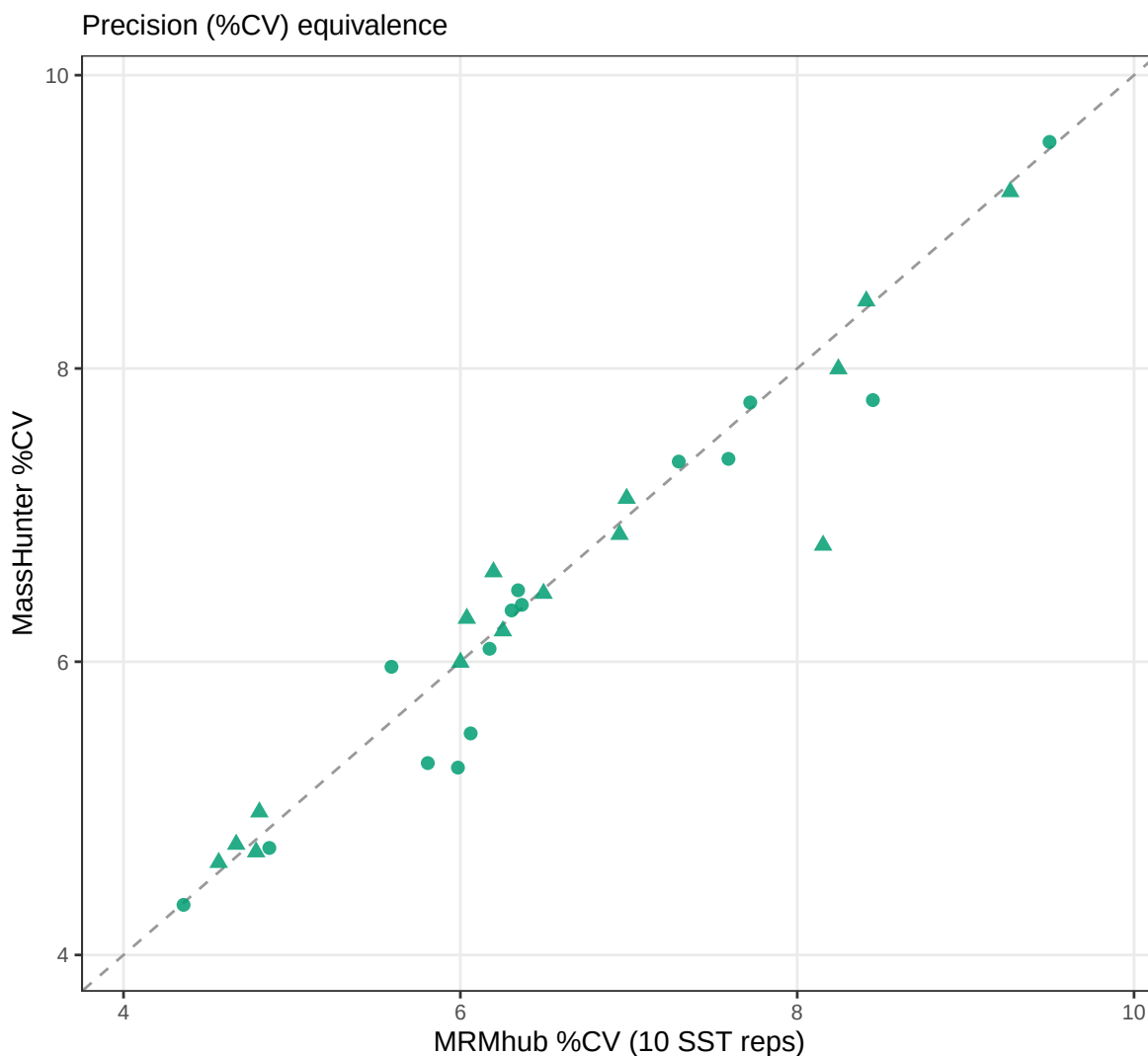



Figure 4: **Precision (%CV) equivalence.** Per-feature %CV over the ten replicate SST injections, MRMhub vs MassHunter; points on the 1:1 line indicate equal precision. Panel B of Figure 7.

4.3 Bland–Altman

Percent difference ($\text{MassHunter} - \text{MRMhub}$) against the mean area, excluding SST and near-noise peaks (mean area $< 10^3$). The three analytes highlighted here have the largest real-sample differences and their chromatograms are shown below.

```
highlight_analytes <- c(
  "21-deoxycortisol",
  "11-deoxycortisol", "Aldosterone")
```

```

hl_cols <- setNames(c("#D81B60", "#1E88E5", "#8E44AD"), highlight_analytes)

d_ba <- d_peakcmp_reint |>
  filter(qc_type != "SST") |>
  mutate(
    mean_area = (area_mrm + area_mh) / 2,
    pct_diff = (area_mh - area_mrm) / mean_area * 100,
    hl = factor(
      if_else(
        feature_id %in% highlight_analytes,
        feature_id, NA_character_),
      levels = highlight_analytes)) |>
  filter(mean_area >= 1e3)
ba_bias <- mean(d_ba$pct_diff)
ba_loa <- ba_bias + c(-1.96, 1.96) * sd(d_ba$pct_diff)
ba_xr <- max(d_ba$mean_area)
p_ba <- d_ba |>
  ggplot(aes(mean_area, pct_diff)) +
  geom_hline(yintercept = 0, colour = "grey70") +
  geom_hline(yintercept = ba_bias, colour = "grey20") +
  geom_hline(yintercept = ba_loa, linetype = "dashed", colour = "grey50") +
  geom_point(
    data = ~ filter(.x, is.na(hl)),
    colour = "grey80", size = 0.9, alpha = 0.6) +
  geom_point(
    data = ~ filter(.x, !is.na(hl)),
    aes(colour = hl), size = 1.7, alpha = 0.9) +
  # Ring the peaks MassHunter integrated manually (MI flag).
  geom_point(
    data = ~ filter(.x, mi_mh),
    aes(alpha = "Manual integration (MassHunter)"), shape = 1,
    colour = "grey15", size = 2.0, stroke = 0.5) +
  annotate(
    "text",
    x = ba_xr, y = ba_bias, hjust = 1, vjust = -0.6, size = 2.1,
    colour = "grey20", label = sprintf("mean bias %.1f%%", ba_bias)) +
  annotate(
    "text",
    x = ba_xr, y = ba_loa[2], hjust = 1, vjust = 1.4, size = 2,
    colour = "grey45", label = "+95% LoA") +
  annotate(
    "text",

```

```

    x = ba_xr, y = ba_loa[1], hjust = 1, vjust = -0.7, size = 2,
    colour = "grey45", label = "-95% LoA") +
scale_colour_manual(
  values = hl_cols,
  name = "Highlighted analyte", na.translate = FALSE) +
scale_alpha_manual(
  name = NULL,
  values = c("Manual integration (MassHunter)" = 1)) +
scale_x_log10(limits = c(1e3, NA)) +
coord_cartesian(ylim = c(-20, 20)) +
guides(
  alpha = guide_legend(
    order = 2,
    override.aes = list(
      shape = 1,
      colour = "grey15", size = 2.0, stroke = 0.5))) +
labs(
  x = "Mean peak area of the two methods ( $\geq 10^3$ )",
  y = "Percent difference (MH - MRMhub) %",
  title = sprintf(
    "Bland-Altman (excl. SST, area  $\geq 10^3$ ): mean bias %.1f%%, 95%% LoA [%.0f, %.0f]%%",
    ba_bias, ba_loa[1], ba_loa[2]))
p_ba

```

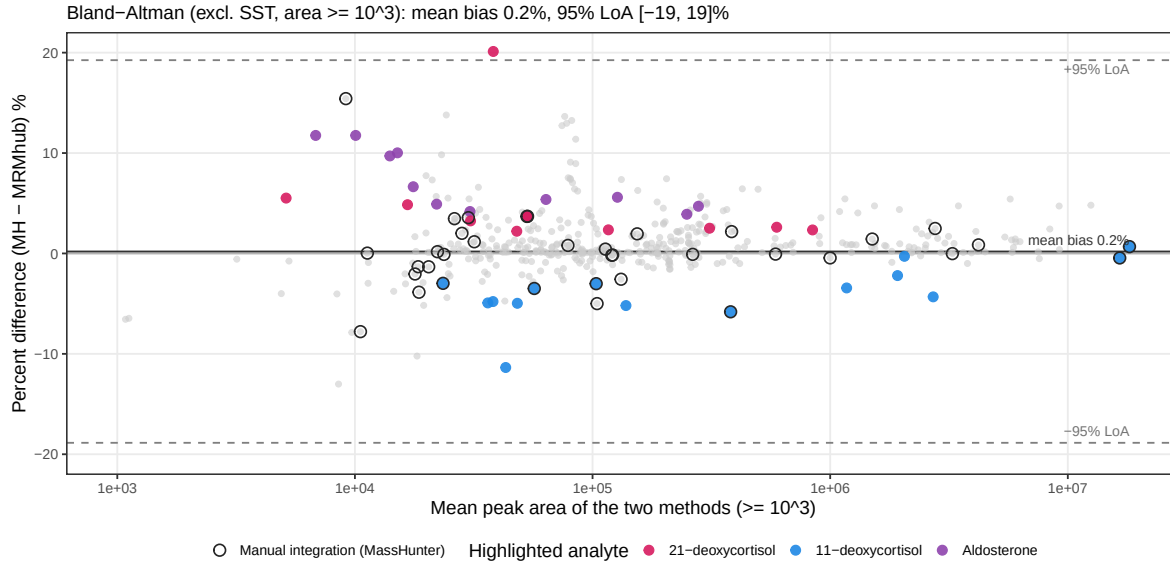


Figure 5: **Bland-Altman of real-sample differences.** Percent difference (MassHunter – MRMhub) against the mean peak area (excl. SST; mean area 10^3); the three largest-difference analytes are highlighted and open grey rings mark manual MassHunter integrations. Panel C of Figure 7.

4.4 Example chromatograms

The largest real-sample area differences occur where a peak sits on an elevated baseline or beside a co-eluting neighbour, so the integration boundaries — and hence the areas — legitimately differ between the two integrators. The candidate table ranks these cases; the panel crops the three chosen examples (one per highlighted analyte) from the MRMhub `by_transition` chromatograms.

```
# Panel-D chromatograms are the pre-generated crops (see the INTEGRATOR section);
# subtitles carry the data-derived area difference for each example.
chrom_examples <- tibble::tribble(
  ~name,
  ~feature_id, ~analysis_id, "fig1d_21-deoxycortisol", "21-deoxycortisol",
  "SKML2025_6B_P1", "fig1d_11-deoxycortisol", "11-deoxycortisol",
  "SKML2026_2B_P1", "fig1d_Aldosterone", "Aldosterone", "SKML2026_2A_P1")

chrom_meta <- chrom_examples |>
  left_join(
    distinct(d_peakcmp_reint, feature_id, analysis_id, area_diff_pct),
    by = c("feature_id", "analysis_id")) |>
  mutate(
    subtitle = sprintf(
```

Table 3: **Largest area-difference candidates.** Peaks ranked by absolute MRMhub–MassHunter area difference, from which the example chromatograms are chosen.

```
chrom_candidates <- d_peakcmp_reint |>
  filter(
    feature_type == "Quantifier",
    qc_type %in% c("CAL", "HQC", "LQC", "SPL", "SST"), area_mrm >= 20000) |>
  arrange(desc(abs(area_diff_pct))) |>
  transmute(
    feature_id,
    analysis_id, qc_type, area_mrm = round(area_mrm),
    area_mh = round(area_mh), area_diff_pct = round(area_diff_pct, 1)) |>
  head(10)
chrom_candidates
#> # A tibble: 10 x 6
#>   feature_id      analysis_id qc_type area_mrm area_mh area_diff_pct
#>   <chr>          <chr>      <chr>    <dbl>    <dbl>         <dbl>
#> 1 21-deoxycortisol SKML2025_6B_P1 SPL      34372    42059          22.4
#> 2 17-hydroxyprogesterone SKML2025_6B_P1 SPL      76446    59981         -21.5
#> 3 11-deoxycortisol SKML2026_2B_P1 SPL      45558    40661         -10.7
#> 4 Dihydrotestosterone Cal D_P1      CAL      22013    24289          10.3
#> 5 DHEA            20260212_SST03 SST      42897    38529         -10.2
#> 6 DHEA            Cal D_P1      CAL      81165    88766           9.4
#> 7 11-deoxycortisol 20260212_SST07 SST     1285343 1389585           8.1
#> 8 Androstenedione Cal A_P1      CAL      76008    81960           7.8
#> 9 DHEA            Cal E_P1      CAL     197230 212468           7.7
#> 10 17-hydroxyprogesterone SKML2026_2Ab_P1 SPL     127355 137090           7.6
```

```

"%s (dArea %+.1f%%)",
  str_remove(analysis_id, "_P1$"), area_diff_pct))
chrom_plots <- Map(
  chrom_gg,
  chrom_meta$name, chrom_meta$feature_id, chrom_meta$subtitle)
p_chrom <- wrap_elements(full = wrap_plots(chrom_plots, nrow = 1))
p_chrom

```

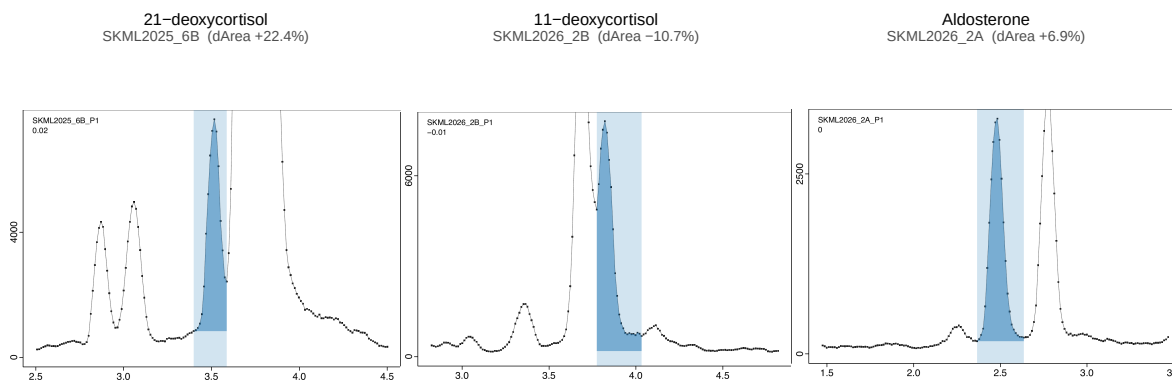


Figure 6: **Example chromatograms.** MRMhub chromatograms for the three highlighted analytes, where the integration boundaries — and hence the areas — legitimately differ between the two integrators. Panel D of Figure 7.

4.5 Figure 1 — manuscript figure

The four panels above are composed into the publication figure (**Figure 1**) and saved as a vector PDF and a 300-dpi PNG (Nature 180 mm width).

```

row_ab <- (p_corr + p_cv) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
fig1 <- row_ab /
  p_ba /
  p_chrom +
  plot_layout(heights = c(1, 0.72, 0.58)) +
  plot_annotation(tag_levels = "A")

save_fig(fig1, "fig1_integration_correspondence", height = 230)
fig1

```

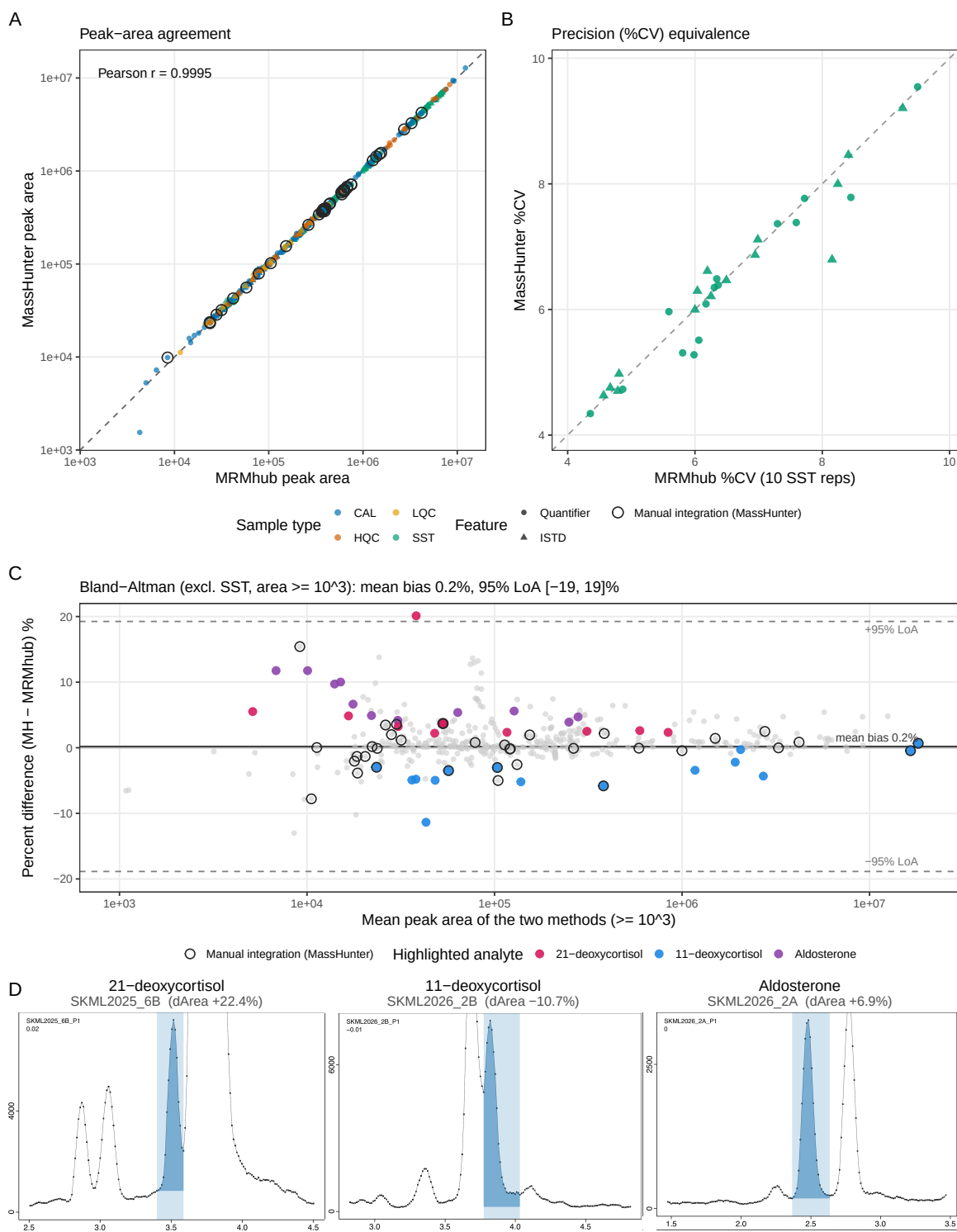


Figure 7: **MRMhub vs MassHunter — peak integration.** Composite of the four panels above; reproduced as the manuscript's **Figure 1**. (A) Peak-area agreement, (B) precision (%CV) equivalence, (C) Bland-Altman of real-sample differences, (D) example chromatograms. Exported to `output/fig1_integration_correspondence.{pdf,png}` (180 mm).

5 MRMhub vs MassHunter — Quantification

Figure 12 compares the two pipelines at the quantification stage. To isolate the post-processing (calibration + quantification) from the peak integration, we also feed **MassHunter's own peak areas through the MRMhub pipeline** — identical areas, identical calibration model — so any concentration difference is due to post-processing alone.

```
mh_area <- mh_data |>
  transmute(analysis_id, feature_name = feature_id, area_mh)
long_mh <- read_csv(long_combined_path, show_col_types = FALSE) |>
  mutate(.aid = str_remove(raw_data_filename, "[.]mzML$")) |>
  inner_join(mh_area, by = c(".aid" = "analysis_id", "feature_name")) |>
  mutate(area = area_mh) |>
  select(-.aid, -area_mh)

tmp_long <- tempfile(fileext = ".csv")
write_csv(long_mh, tmp_long)

mexp_mh <- MRMhubExperiment(title = "MassHunter areas via MRMhub pipeline")
mexp_mh <- import_data_mrmhub(
  mexp_mh,
  path = tmp_long,
  import_metadata = TRUE)
#> v Imported 30 analyses with 30 features.
#> i feature_area selected as default feature intensity. Modify with `set_intensity_var()`.
#> Warning: ! Unrecognized qc_type value(s): "BLK".
#> i These are not standard QC types and may be dropped from QC metrics and plots
#>   that expect them.
#> i Standard values: "SBLK", "TBLK", "UBLK", "HQC", "MQC", "LQC", "QC", "PBLK",
#>   "CAL", "EQA", "PQC", "TQC", "BQC", "RQC", "EQC", "NIST", "LTR", "SPL", "SST",
#>   and "MBLK" ("Sample" is an alias for "SPL").
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
mexp_mh <- import_metadata_msorganiser(
  mexp_mh,
  path = meta_path,
  excl_unmatched_analyses = TRUE, ignore_warnings = TRUE)
#> Found no errors, 1 warning, and 2 notes in the metadata.
#> -----
#>   Type  Table      Column      Issue                                     Count
#> 1 W*    Analyses analysis_id Analyses not in analysis data             2
#> 2 N     Analyses sample_id   Not defined for all analyses          16
```



```

#> 3 N      Features analyte_id Not defined for all features      15
#>
#> -----
#> E = Error, W = Warning, W* = Suppressed Warning, N = Note
#> -----
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
#> v Internal Standard metadata associated with 15 ISTDs.
#> v QC concentration metadata associated with 13 annotated samples and 15 annotated analytes.
mexp_mh <- normalize_by_istd(mexp_mh)
#> Warning: ! 2 feature values could not be ISTD-normalized because the internal standard
#>   had zero intensity and were set to NA.
#> i Affected analyses: "Blank2_P1" and "Blank3_P1"
#> v 15 features normalized with 15 ISTDs in 30 analyses.
mexp_mh <- calc_calibration_results(
  mexp_mh,
  fit_overwrite = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x")
#> v Calibration curve fits calculated for all 15 quantifier features. Average r-squared: 0.9
mexp_mh <- quantify_by_calibration(
  mexp_mh,
  fit_overwrite = FALSE,
  include_qualifier = FALSE,
  ignore_failed_calibration = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x")
#> v Calibration curve fits calculated for all 15 quantifier features. Average r-squared: 0.9
#> i 80 concentration values fall outside the calibrated range (retained, flagged in feature)
#> v Concentrations calculated for 15 features in 30 analyses.
#> v Concentrations are given in nmol/L.

```

5.1 Calibration fit comparison

Calibration R^2 per analyte: MassHunter's own fit vs the MRMhub full-pipeline fit.

```

mh_own_r2 <- read_csv(
  file.path(d4, "Dataset4_MassHunter-CalibrationFits.csv"),
  show_col_types = FALSE) |>
  transmute(analyte_id, r2_mh_own = r2)
r2_cmp <- get_calibration_metrics(mexp_mh) |>

```

```

filter(is_quantifier) |>
transmute(analyte_id = feature_id, r2_mrmpipe_mharea = r2) |>
left_join(mh_own_r2, by = "analyte_id") |>
left_join(
  get_calibration_metrics(mexp) |>
    filter(is_quantifier) |>
    transmute(analyte_id = feature_id, r2_mrmpipe_mrmarea = r2),
  by = "analyte_id")

chain_lv <- c("MassHunter", "MRMhub")
chain_col <- c(MassHunter = "#D95F0E", MRMhub = "#2C7FB8")
ord_r2 <- r2_cmp |> arrange(r2_mrmpipe_mrmarea) |> pull(analyte_id)

d_r2_long <- r2_cmp |>
  transmute(
    analyte_id,
    MassHunter = r2_mh_own, MRMhub = r2_mrmpipe_mrmarea) |>
  pivot_longer(-analyte_id, names_to = "chain", values_to = "r2") |>
  mutate(
    analyte_id = factor(analyte_id, levels = ord_r2),
    chain = factor(chain, levels = chain_lv))

# Truncate the x-axis just below the dense 0.996-1.000 cluster; flag off-scale rows.
R2_FLOOR <- 0.9955
r2_off <- d_r2_long |>
  filter(r2 < R2_FLOOR) |>
  distinct(analyte_id) |>
  mutate(r2 = R2_FLOOR)
off_txt <- d_r2_long |>
  filter(r2 < R2_FLOOR) |>
  group_by(analyte_id) |>
  summarise(v = min(r2), .groups = "drop") |>
  arrange(v) |>
  mutate(t = sprintf("%s %.3f", analyte_id, v))
p_r2 <- ggplot(d_r2_long, aes(r2, analyte_id)) +
  geom_line(aes(group = analyte_id), colour = "grey75", linewidth = 0.6) +
  geom_point(aes(colour = chain), size = 1.9, alpha = 0.9) +
  geom_text(
    data = r2_off,
    aes(label = "<"), hjust = 0, nudge_x = 0.00004, size = 2.3,
    colour = "grey45") +
  scale_colour_manual(values = chain_col, name = NULL) +

```

```

scale_x_continuous(breaks = c(0.996, 0.997, 0.998, 0.999, 1.000)) +
coord_cartesian(xlim = c(R2_FLOOR, 1.0004)) +
labs(
  x = expression(Calibration ~ R2),
  y = NULL, title = "Calibration curve fit comparison",
  subtitle = paste0(
    "x-axis truncated at ",
    R2_FLOOR, "; off scale (<): ", paste(off_txt$t, collapse = ", ")) +
  theme(
    plot.subtitle = element_text(size = 6.5, colour = "grey30"),
    plot.margin = margin(4, 10, 4, 4), legend.position = "inside",
    legend.position.inside = c(0.015, 0.98),
    legend.justification.inside = c(0, 1),
    legend.background = element_rect(
      fill = alpha("white", 0.7),
      colour = NA),
    legend.key.size = unit(3, "mm"))
p_r2

```

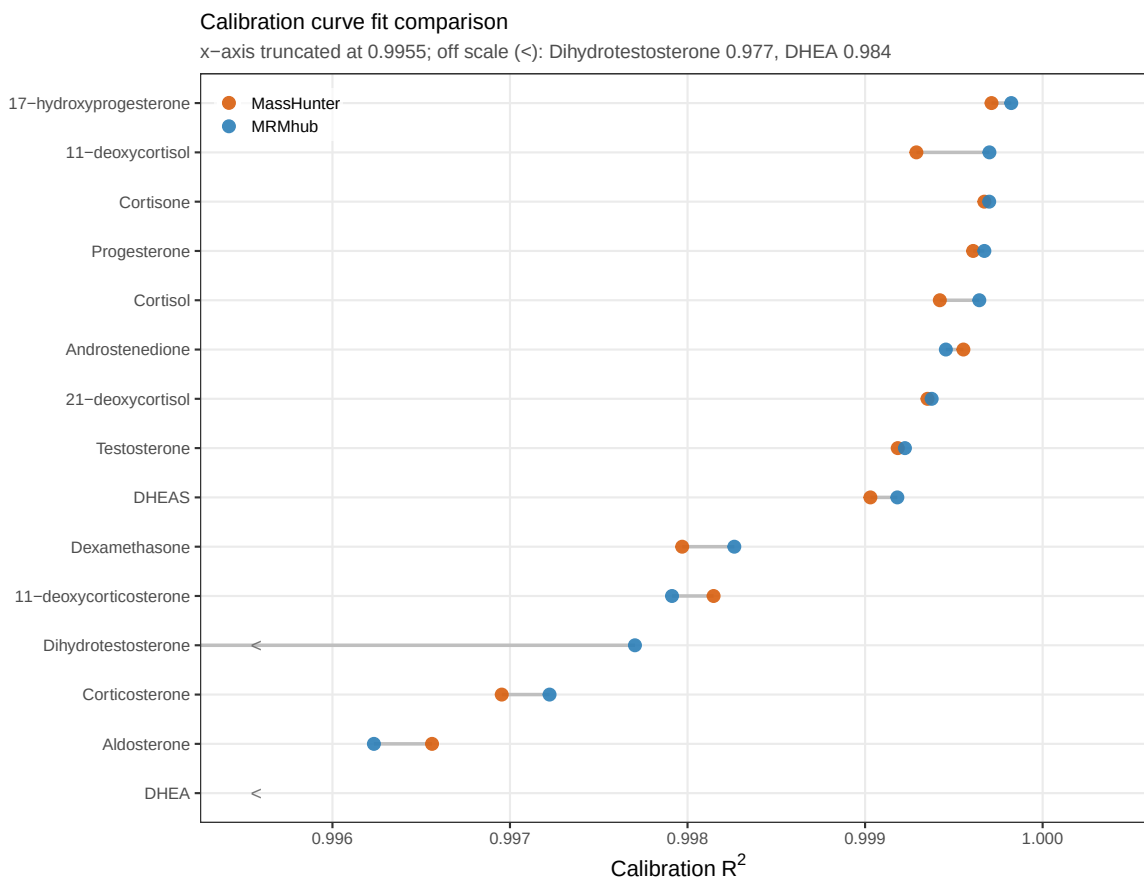


Figure 8: **Calibration-fit comparison.** R² per analyte, MassHunter's own fit vs the MRMhub full-pipeline fit. Panel A of Figure 12.

The same comparison as a table, with **three fits per analyte** (quadratic, $y = a \cdot x^2 + b \cdot x + c$, 1/x weighting): MassHunter's own fit, the MRMhub end-to-end fit (MRMhub-INTEGRATOR areas), and the MRMhub pipeline run on **MassHunter's own peak areas**. The gap between the two MRMhub rows is the peak-integration effect; the distance between *MRMhub (on MH areas)* and *MassHunter* isolates the post-processing (normalization, weighting, regression) alone.

```
library(gt)

# Measured LQC / HQC concentrations (nmol/L) per engine, one column each.
qc_wide <- function(df)
  df |>
    filter(qc_type %in% c("LQC", "HQC")) |>
    pivot_longer(
```

```

      !c(analysis_id, qc_type),
      names_to = "analyte_id", values_to = "conc") |>
select(analyte_id, qc_type, conc) |>
pivot_wider(names_from = qc_type, values_from = conc)

tmp_mhconc <- tempfile(fileext = ".csv")
save_dataset_csv(
  mexp_mh,
  path = tmp_mhconc,
  variable = "conc",
  add_qctype = TRUE)
#> v Concentration values for 30 analyses and 15 features have been exported to '/var/folders
conc_e2e <- qc_wide(d_conc_raw)
conc_onmh <- qc_wide(read_csv(tmp_mhconc, show_col_types = FALSE))
conc_mhun <- mh_data |>
  filter(analysis_id %in% c("QC_Low1_P1", "QC_High1_P1")) |>
  transmute(
    analyte_id = feature_id,
    qc_type = if_else(analysis_id == "QC_Low1_P1", "LQC", "HQC"),
    conc = conc_mh) |>
  pivot_wider(names_from = qc_type, values_from = conc)

# mrmhub reports coefficients intercept-first (coef_a = x^0 ... coef_c = x^2);
# remap to MassHunter's descending order (a = x^2, b = x, c = intercept).
mh_own <- read_csv(
  file.path(d4, "Dataset4_MassHunter-CalibrationFits.csv"),
  show_col_types = FALSE) |>
  transmute(
    analyte_id,
    method = "MassHunter", a = coef_a, b = coef_b, c = coef_c, r2 = r2) |>
  left_join(conc_mhun, by = "analyte_id")
mrm_mh <- get_calibration_metrics(mexp_mh) |>
  filter(is_quantifier) |>
  transmute(
    analyte_id = feature_id,
    method = "MRMhub (on MH areas)", a = coef_c, b = coef_b, c = coef_a,
    r2 = r2) |>
  left_join(conc_onmh, by = "analyte_id")
mrm_e2e <- get_calibration_metrics(mexp) |>
  filter(is_quantifier) |>
  transmute(
    analyte_id = feature_id,

```

```

    method = "MRMhub (end-to-end)", a = coef_c, b = coef_b, c = coef_a,
    r2 = r2) |>
left_join(conc_e2e, by = "analyte_id")

cal_lv <- c("MassHunter", "MRMhub (on MH areas)", "MRMhub (end-to-end)")
fmt_conc <- function(x)
  ifelse(
    is.na(x),
    "_",
    formatC(
      signif(x, 4),
      format = "fg", big.mark = ",", drop0trailing = TRUE))
cal_coef_tbl <- bind_rows(mh_own, mrm_mh, mrm_e2e) |>
mutate(method = factor(method, levels = cal_lv)) |>
arrange(analyte_id, method) |>
transmute(
  Analyte = if_else(as.character(method) == cal_lv[1], analyte_id, ""),
  Method = as.character(method), `a (x2)` = sprintf("%.4g", a),
  `b (x)` = sprintf("%.4g", b), `c` = sprintf("%.4g", c),
  `R2` = sprintf("%.5f", r2), `LQC Concentration` = fmt_conc(LQC),
  `HQC Concentration` = fmt_conc(HQC))

gt(cal_coef_tbl) |>
cols_align(
  "right",
  columns = c(
    "a (x2)",
    "b (x)", "c", "R2", "LQC Concentration", "HQC Concentration")) |>
tab_style(
  cell_fill(color = "#DEEBF7"),
  cells_body(rows = Method == "MRMhub (on MH areas)")) |>
tab_style(
  list(cell_fill(color = "#2C7FB8"), cell_text(color = "white")),
  cells_body(rows = Method == "MRMhub (end-to-end)")) |>
tab_options(table.font.size = px(12), data_row.padding = px(2))

```

5.2 Concentration equivalence on identical areas

Feeding MassHunter areas through the MRMhub pipeline reproduces MassHunter's end-to-end concentrations almost exactly — the post-processing steps are equivalent.

Table 4: **Calibration fit per analyte.** Coefficients of $y = a \cdot x^2 + b \cdot x + c$, R^2 , and measured LQC/HQC concentrations (nmol/L) for three fits — MassHunter’s own, MRMhub on MassHunter areas, and MRMhub end-to-end (blue text). The two MRMhub rows differ only by peak integration; MRMhub-on-MH-areas vs MassHunter isolates post-processing.

Analyte	Method	a (x^2)	b (x)	c	R^2	LQC Concentration	HQC
11-deoxycorticosterone	MassHunter	0.01256	1.471	0.04589	0.99815	0.6529	
	MRMhub (on MH areas)	0.01256	1.471	0.0459	0.99815	0.6529	
	MRMhub (end-to-end)	0.01313	1.427	0.04811	0.99791	0.6723	
11-deoxycortisol	MassHunter	0.005944	0.9796	0.06228	0.99929	1.458	
	MRMhub (on MH areas)	0.005944	0.9796	0.06228	0.99929	1.458	
	MRMhub (end-to-end)	0.005598	1.022	0.0644	0.99970	1.451	
17-hydroxyprogesterone	MassHunter	0.001839	0.9685	0.03238	0.99971	1.488	
	MRMhub (on MH areas)	0.001839	0.9685	0.03238	0.99971	1.488	
	MRMhub (end-to-end)	0.001363	0.9671	0.03846	0.99982	1.513	
21-deoxycortisol	MassHunter	0.006042	0.7758	0.008574	0.99935	1.698	
	MRMhub (on MH areas)	0.006042	0.7758	0.008574	0.99935	1.697	
	MRMhub (end-to-end)	0.005988	0.7606	0.002312	0.99938	1.684	
Aldosterone	MassHunter	0.01507	0.5417	0.1124	0.99656	0.9199	
	MRMhub (on MH areas)	0.01507	0.5417	0.1124	0.99656	0.9199	
	MRMhub (end-to-end)	0.01406	0.518	0.08996	0.99623	0.9878	
Androstenedione	MassHunter	0.005103	0.6132	0.04322	0.99955	1.81	
	MRMhub (on MH areas)	0.005103	0.6132	0.04322	0.99955	1.81	
	MRMhub (end-to-end)	0.004824	0.6072	0.0427	0.99945	1.842	
Corticosterone	MassHunter	-6.449e-05	0.4008	-0.01636	0.99695	3.758	
	MRMhub (on MH areas)	-6.45e-05	0.4008	-0.01636	0.99695	3.757	
	MRMhub (end-to-end)	-6.664e-05	0.4109	-0.01616	0.99722	3.601	
Cortisol	MassHunter	1.227e-05	0.1237	-0.06246	0.99942	24.37	
	MRMhub (on MH areas)	1.227e-05	0.1237	-0.06246	0.99942	24.37	
	MRMhub (end-to-end)	1.526e-05	0.1212	-0.03548	0.99964	24.89	
Cortisone	MassHunter	-0.0003965	0.4315	-0.1158	0.99967	6.238	
	MRMhub (on MH areas)	-0.0003965	0.4315	-0.1158	0.99967	6.238	
	MRMhub (end-to-end)	-0.0003793	0.428	-0.09925	0.99970	6.269	
DHEA	MassHunter	-3.25e-05	0.03091	0.1772	0.98415	8.355	
	MRMhub (on MH areas)	-3.25e-05	0.03091	0.1772	0.98415	8.355	
	MRMhub (end-to-end)	-4.076e-05	0.03256	0.1832	0.98458	8.937	
DHEAS	MassHunter	-7.313e-09	0.003939	-0.0933	0.99903	524.2	
	MRMhub (on MH areas)	-7.313e-09	0.003939	-0.0933	0.99903	524.2	
	MRMhub (end-to-end)	-1.016e-08	0.003949	-0.09189	0.99918	521.2	
Dexamethasone	MassHunter	-1.546e-05	0.1036	-0.01331	0.99797	6.046	
	MRMhub (on MH areas)	-1.546e-05	0.1036	-0.01331	0.99797	6.046	
	MRMhub (end-to-end)	-1.714e-05	0.1036	-0.007976	0.99826	5.983	
Dihydrotestosterone	MassHunter	-0.03613	0.5642	-0.1789	0.97656	1.224	
	MRMhub (on MH areas)	-0.03613	0.5642	-0.1789	0.97656	1.224	
	MRMhub (end-to-end)	-0.01988	0.4645	-0.07173	0.99770	1.212	
Progesterone	MassHunter	-0.002326	0.9786	0.03361	0.99961	1.349	
	MRMhub (on MH areas)	-0.002326	0.9786	0.03361	0.99961	1.349	
	MRMhub (end-to-end)	-0.001442	0.9495	0.05202	0.99967	1.322	
Testosterone	MassHunter	0.003025	1.22	0.07571	0.99918	0.87	
	MRMhub (on MH areas)	0.003025	1.22	0.07571	0.99918	0.87	
	MRMhub (end-to-end)	0.002567	1.216	0.0762	0.99922	0.8558	

```

tmp_conc <- tempfile(fileext = ".csv")
save_dataset_csv(
  mexp_mh,
  path = tmp_conc,
  variable = "conc",
  add_qctype = TRUE)
#> v Concentration values for 30 analyses and 15 features have been exported to '/var/folders
d_qonly <- read_csv(tmp_conc, show_col_types = FALSE) |>
  pivot_longer(
    !c(analysis_id, qc_type),
    names_to = "analyte_id", values_to = "conc_mrmpipe") |>
  left_join(
    mh_data |> select(analysis_id, analyte_id = feature_id, conc_mh),
    by = c("analysis_id", "analyte_id")) |>
  filter(!is.na(conc_mrmpipe), !is.na(conc_mh)) |>
  filter(!qc_type %in% c("SBLK", "PBLK")) |>
  mutate(qc_type = factor(qc_type, levels = names(smp_cols)))
posq <- d_qonly |> filter(conc_mrmpipe > 0, conc_mh > 0)
q_r <- cor(log10(posq$conc_mrmpipe), log10(posq$conc_mh))
axq <- range(c(posq$conc_mrmpipe, posq$conc_mh))
p_qonly <- posq |>
  ggplot(aes(conc_mh, conc_mrmpipe, colour = qc_type)) +
  geom_abline(
    slope = 1,
    intercept = 0, linetype = "dashed", colour = "grey40") +
  geom_point(size = 1.1, alpha = 0.6) +
  annotate(
    "text",
    x = axq[1], y = axq[2], hjust = 0, vjust = 1, size = 2.4,
    label = sprintf("Pearson r = %.5f", q_r)) +
  scale_x_log10(limits = axq) +
  scale_y_log10(limits = axq) +
  scale_colour_manual(
    values = smp_cols,
    drop = TRUE, name = "Sample type") +
  coord_equal() +
  labs(
    x = "MassHunter End-to-End",
    y = "MRMhub-QUANT on MH areas",
    title = "Concentration equivalence (same areas)") +
  theme(
    legend.position = "inside",

```



```
legend.position.inside = c(0.98, 0.02),  
legend.justification.inside = c(1, 0),  
legend.background = element_rect(  
  fill = alpha("white", 0.7),  
  colour = NA),  
legend.key.size = unit(2.8, "mm"),  
legend.title = element_text(size = 6.5),  
legend.text = element_text(size = 6))  
p_qonly
```

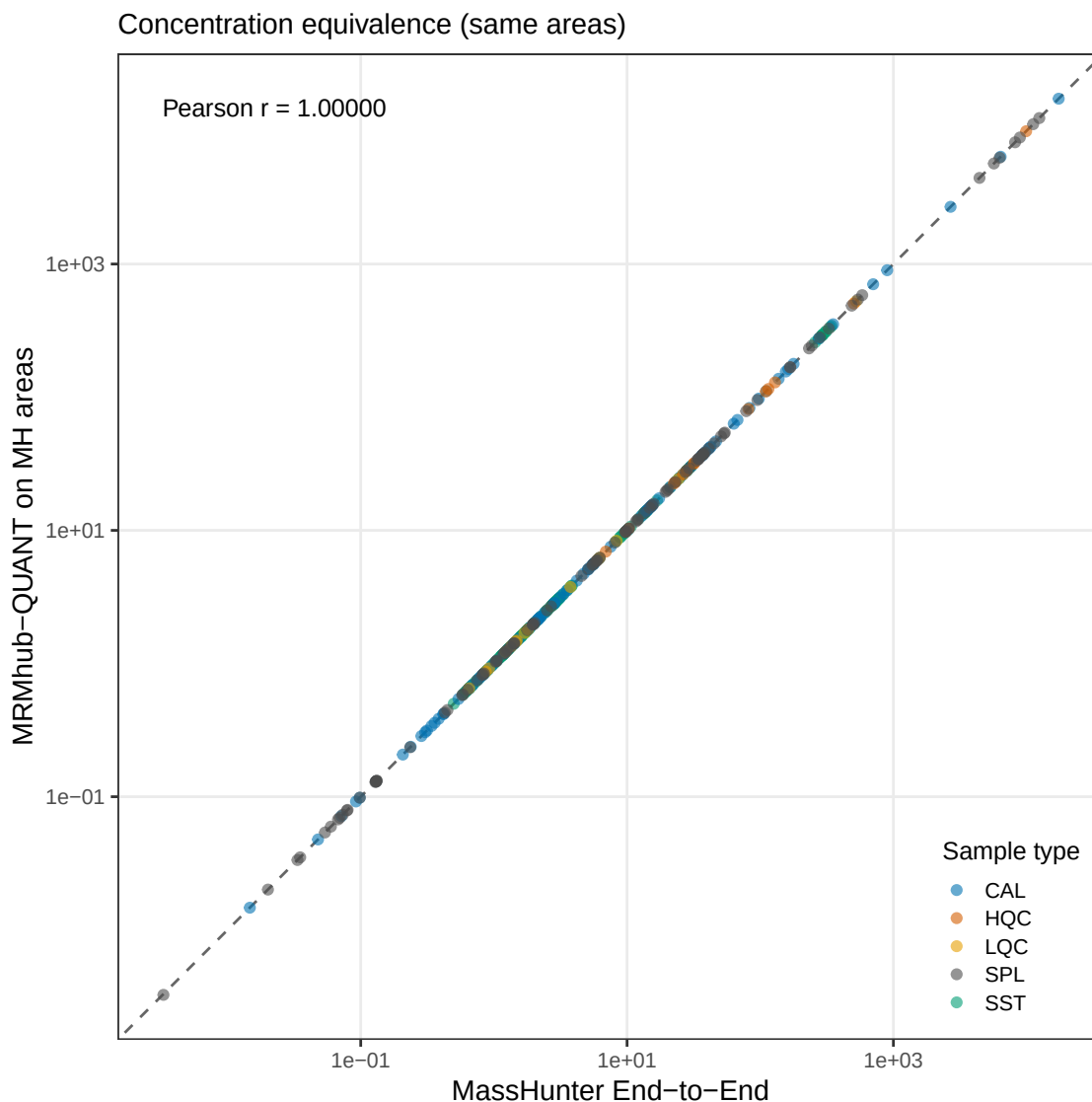


Figure 9: **Concentration equivalence on identical areas.** MassHunter areas run through the MRMhub pipeline vs MassHunter end-to-end, isolating post-processing from integration. Panel B of Figure 12.

5.3 Full end-to-end concentration agreement

Independent end-to-end concentrations (each method's own integration and calibration) for the LQC, HQC and SPL samples.

```
d_full <- d_conc_raw |>
  filter(qc_type %in% c("LQC", "HQC", "SPL")) |>
```

```

pivot_longer(
  !c("analysis_id", "qc_type"),
  names_to = "analyte_id", values_to = "conc") |>
left_join(
  mh_data |> select(analysis_id, analyte_id = feature_id, conc_mh),
  by = c("analysis_id", "analyte_id")) |>
left_join(
  select(spl_area_excl, analysis_id, analyte_id, excl_mrm, excl_mh),
  by = c("analysis_id", "analyte_id")) |>
mutate(
  conc = if_else(coalesce(excl_mrm, FALSE), NA_real_, conc),
  conc_mh = if_else(coalesce(excl_mh, FALSE), NA_real_, conc_mh)) |>
select(-excl_mrm, -excl_mh) |>
filter(!is.na(conc), !is.na(conc_mh), conc > 0, conc_mh > 0) |>
mutate(qc_type = factor(qc_type, levels = names(smp_cols)))
f_r <- cor(log10(d_full$conc), log10(d_full$conc_mh))
axf <- range(c(d_full$conc, d_full$conc_mh))
p_full <- d_full |>
ggplot(aes(conc_mh, conc, colour = qc_type)) +
geom_abline(
  slope = 1,
  intercept = 0, linetype = "dashed", colour = "grey40") +
geom_point(size = 1.3, alpha = 0.7) +
annotate(
  "text",
  x = axf[1], y = axf[2], hjust = 0, vjust = 1, size = 2.4,
  label = sprintf("Pearson r = %.4f", f_r)) +
scale_x_log10(limits = axf) +
scale_y_log10(limits = axf) +
scale_colour_manual(
  values = smp_cols,
  drop = TRUE, name = "Sample type") +
coord_equal() +
labs(
  x = "MassHunter (end-to-end)",
  y = "MRMhub (end-to-end)",
  title = "Concentration agreement (full pipeline)") +
theme(legend.position = "none")
p_full

```

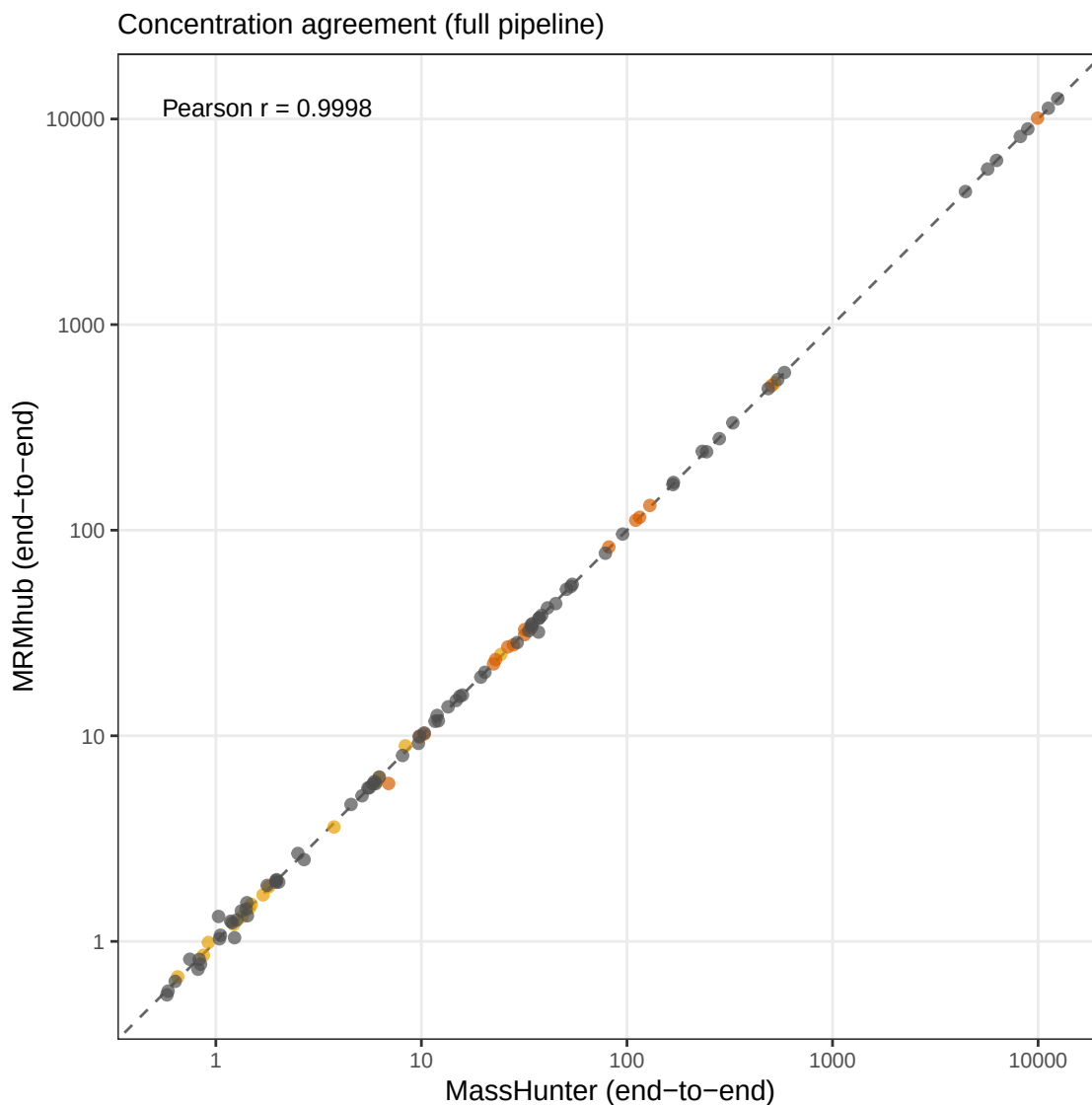


Figure 10: **Full end-to-end concentration agreement.** Each method's own integration and calibration, for the LQC, HQC and SPL samples. Panel C of Figure 12.

5.4 EQA accuracy against SKML reference values

Per-analyte percent bias against the SKML EQA reference concentrations, one point per EQA sample per method (dashed lines: $\pm 15\%$). The reference concentrations are read from the metadata workbook's **QC Concentrations** sheet (imported above with the rest of the metadata), keyed by sample and analyte. DHEA and DHEAS are excluded (large biases that would compress the other panels).

```

# EQA target concentrations come from the metadata's `QC Concentrations` sheet
# (imported above), keyed by `sample_id` + `analyte_id`. They surface via the
# `annot_qcconcentrations` slot; DHEAS is already stored in nmol/L there.
eqa_target <- mexp@annot_qcconcentrations |>
  filter(str_starts(sample_id, "SKML")) |>
  transmute(sample_id, analyte_id, target = concentration)
# analysis_id -> sample_id map (groups the `...b` replicates onto one EQA sample).
sid_map <- get_analyticaldata(mexp, annotated = TRUE) |>
  distinct(analysis_id, sample_id)
d_eqa <- d_conc_raw |>
  filter(qc_type == "SPL") |>
  pivot_longer(
    !c("analysis_id", "qc_type"),
    names_to = "analyte_id", values_to = "conc_mrm") |>
  left_join(
    mh_data |> select(analysis_id, analyte_id = feature_id, conc_mh),
    by = c("analysis_id", "analyte_id")) |>
  left_join(
    select(spl_area_excl, analysis_id, analyte_id, excl_mrm, excl_mh),
    by = c("analysis_id", "analyte_id")) |>
  mutate(
    conc_mrm = if_else(coalesce(excl_mrm, FALSE), NA_real_, conc_mrm),
    conc_mh = if_else(coalesce(excl_mh, FALSE), NA_real_, conc_mh)) |>
  left_join(sid_map, by = "analysis_id") |>
  inner_join(eqa_target, by = c("sample_id", "analyte_id")) |>
  # Compact axis label, e.g. SKML2026_1A -> 2026.1A.
  mutate(
    sample_eqa = str_replace(
      str_remove(sample_id, "^SKML"),
      "^[0-9]{4})_", "\\1.") |>
  transmute(
    analyte_id,
    sample_eqa, analysis_id, target, conc_mrm, conc_mh,
    bias_mrmhub = (conc_mrm - target) / target * 100,
    bias_masshunter = (conc_mh - target) / target * 100) |>
  filter(!is.na(conc_mrm) & is.na(conc_mh))
p_eqa <- d_eqa |>
  filter(!analyte_id %in% c("DHEA", "DHEAS")) |>
  pivot_longer(
    c(bias_mrmhub, bias_masshunter),
    names_to = "method", values_to = "bias") |>
  filter(!is.na(bias)) |>

```

```

mutate(
  method = recode(
    method,
    bias_mrmhub = "MRMhub", bias_masshunter = "MassHunter"),
  method = factor(method, levels = c("MRMhub", "MassHunter"))) |>
ggplot(aes(sample_eqa, bias, colour = method, shape = method)) +
geom_hline(yintercept = 0, colour = "grey40") +
geom_hline(
  yintercept = c(-15, 15),
  linetype = "dashed", colour = "grey75") +
geom_point(
  size = 1.5,
  alpha = 0.85, position = position_dodge(width = 0.4)) +
scale_colour_manual(values = col_method, name = "Method") +
scale_shape_manual(
  values = c(MRMhub = 16, MassHunter = 17),
  name = "Method") +
facet_wrap(vars(analyte_id), scales = "free_y", ncol = 4) +
labs(
  x = NULL,
  y = "Bias vs EQA reference (%)",
  title = "EQA accuracy - bias per analyte (excl. DHEA, DHEAS)") +
theme(
  panel.grid.minor = element_blank(),
  strip.text = element_text(size = 6),
  axis.text.x = element_text(angle = 45, hjust = 1, size = 5),
  axis.text.y = element_text(size = 5.2))
p_eqa

```

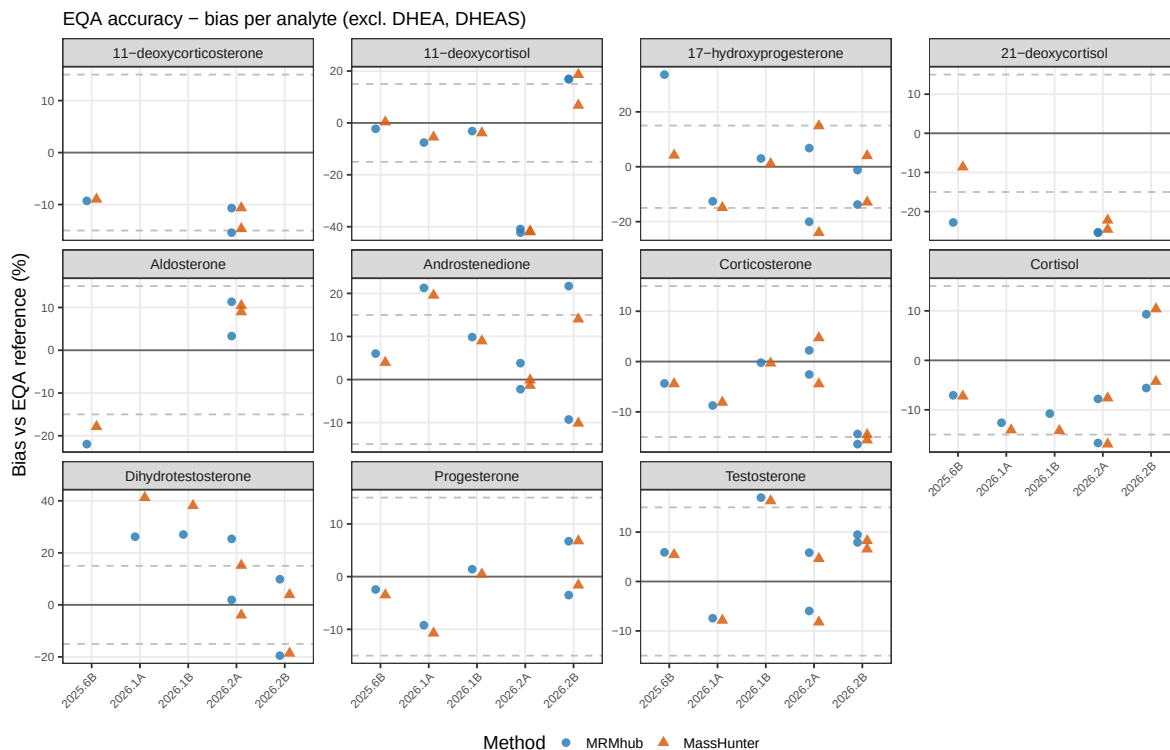


Figure 11: **EQA accuracy against SKML reference values.** Per-analyte percent bias against the SKML reference concentrations (± 15 % dashed), MRMhub vs MassHunter, one point per EQA sample per method (DHEA and DHEAS excluded). Panel D of Figure 12.

5.5 Figure 2 — manuscript figure

The four panels above are composed into the publication figure (**Figure 2**).

```
# Middle row (B, C) uses coord_equal(); give it enough height that the two square
# panels grow to fill the full width, so their L/R borders align with A and D.
fig2 <- p_r2 /
  (p_qonly + p_full) /
  p_eqa +
  plot_layout(heights = c(0.68, 1.3, 1.2)) +
  plot_annotation(tag_levels = "A")

save_fig(fig2, "fig2_quantification_comparison", height = 236)
fig2
```

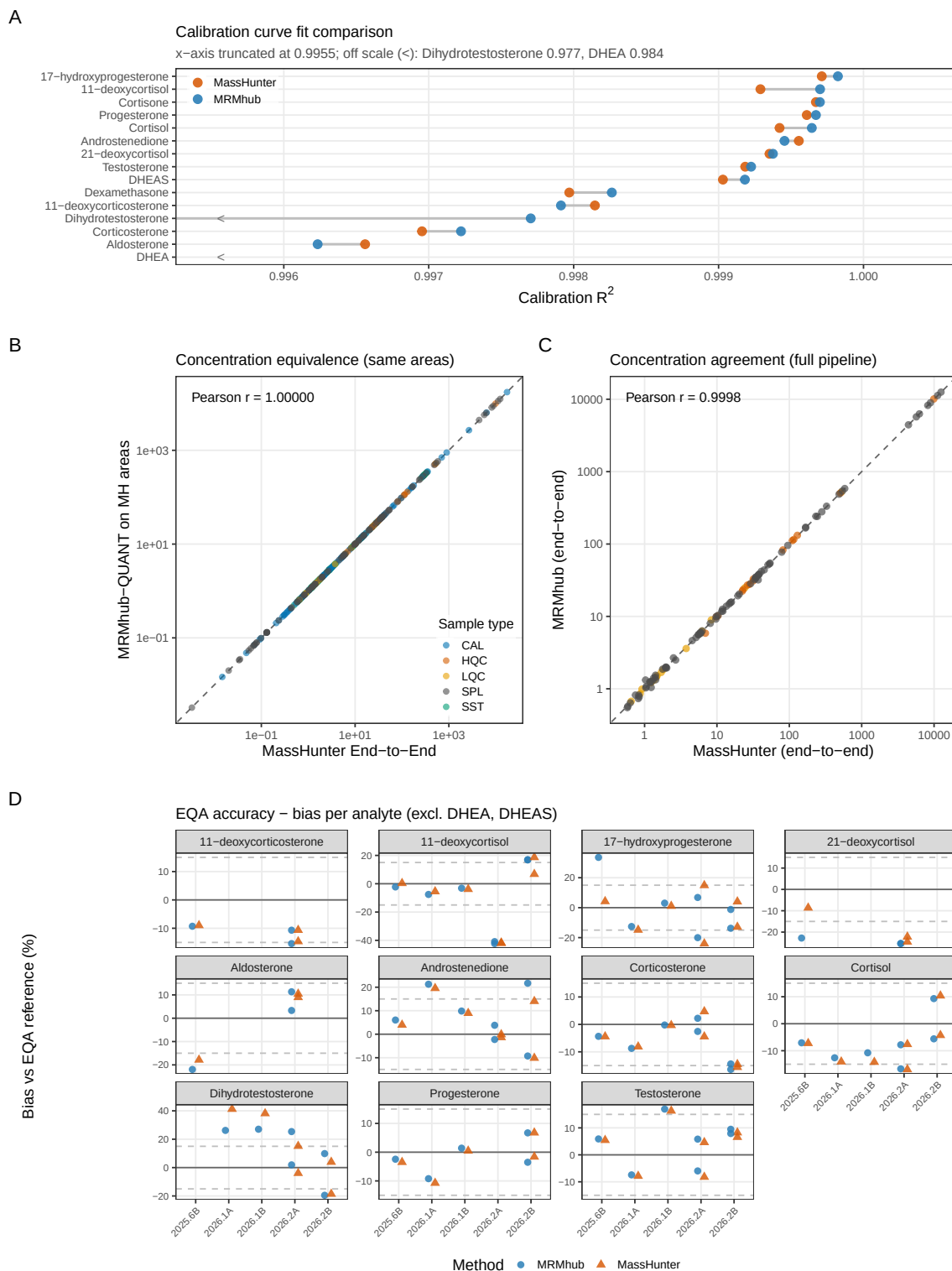


Figure 12: **MRMhub vs MassHunter — quantification.** Composite of the four panels above; reproduced as the manuscript's **Figure 2**. (A) Calibration-fit comparison, (B) concentration equivalence on identical areas, (C) full end-to-end agreement, (D) EQA accuracy. Exported to `output/fig2_quantification_comparison.{pdf,svg}` (180 mm).

References

- Jansen, Rob, Nuthar Jassam, Annette Thomas, Carmen Perich, Pilar Fernandez-Calle, Ana Paula Faria, Helena Correia, et al. 2014. “A Category 1 EQA Scheme for Comparison of Laboratory Performance and Method Performance: An International Pilot Study in the Framework of the Calibration 2000 Project.” *Clinica Chimica Acta* 432 (May): 90–98. <https://doi.org/10.1016/j.cca.2013.11.003>.